WEEK 1 · LESSON 4 OF 24 · HANDOUT

The Types of AI Problems

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=R9OHn5ZF4Uo — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

Your goal in this lesson: Name the main types of AI problems, recognise each one from a real-world description, and choose the right type for a new problem.

Why problem type comes first

Before choosing a tool or a model, an AI practitioner asks one question: **what kind of output do we need?** The answer decides the data you collect, the method you use and how you measure success. Getting the problem type wrong is one of the most common and expensive mistakes in AI projects.

The main problem types

Type	Output	Real example
Classification	A category (a label from a fixed list)	Is this MoMo transaction *fraud* or *normal*? Is this cassava leaf *healthy*, *mosaic disease* or *brown streak*?
Regression	A number	What will this house in Kicukiro sell for? How many litres of milk will the cooperative collect next week?
Clustering	Groups that the data forms by itself, with no labels given	Group customers by buying habits so you can market to each group
Recommendation	A ranked list of items for a person	Which videos or products to show this user next
Anomaly detection	"This is unusual"	Spot a strange login or a broken machine sensor
Generation	New content: text, images, audio, code	Write a product description; draw an illustration
Sequential decision-making	A series of actions	A robot finding its way; a game-playing agent

Classification vs. regression — the classic pair

The easiest test: **is the answer a category or a quantity?**

- "Will this student pass or fail?" → category → **classification**.
- "What mark will this student get out of 100?" → number → **regression**.

The same situation can be framed either way. A bank can ask "will this loan be repaid — yes or no?" (classification) or "how much of this loan will be repaid?" (regression). The right framing depends on the decision the business must make.

Binary classification has two classes (fraud / normal). **Multi-class** has more than two (three cassava diseases plus healthy).

Supervised, unsupervised — a first look

- If you have **examples with the correct answers** (past transactions marked fraud or normal), the problem is **supervised**.
- If you have data **without answers** and want the computer to find structure (customer groups), it is **unsupervised**.

You will study both in Week 2.

A quick decision guide

1. Do you need new content? → **Generation**.
2. Do you need to choose actions over time? → **Sequential decision-making**.
3. Is the answer a category? → **Classification**.
4. Is it a number? → **Regression**.
5. No answers available, want natural groups? → **Clustering**.
6. Want to flag rare strange cases? → **Anomaly detection**.
7. Ranking items for a person? → **Recommendation**.

Remember this: Always ask "what output do we need?" first. Category means classification, number means regression, and the other types cover groups, rankings, rare events, new content and actions.

Name: _____ Date: _____

Activity

Task 1

Predict tomorrow's maximum temperature in Huye.

Task 2

Decide if an email is spam.

Task 3

Group 10,000 farmers by the crops they grow and their yields.

Task 4

Write a thank-you message for a customer.

Task 5

Find unusual electricity use in a factory.

Task 6

Suggest courses a student might like next.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is the first question to ask before choosing an AI method?
   - A. How many people work on the team
   - B. What kind of output do we need?
   - C. Which programming language is popular
   - D. Which laptop to buy
2. Predicting the selling price of a house is…
   - A. Classification
   - B. Clustering
   - C. Regression
   - D. Generation
3. Deciding whether a cassava leaf is healthy or has one of two diseases is…
   - A. Binary classification
   - B. Multi-class classification
   - C. Regression
   - D. Clustering
4. You have customer data with no labels and want to find natural groups. This is…
   - A. Classification
   - B. Generation
   - C. Regression
   - D. Clustering
5. Writing a product description automatically is…
   - A. Generation
   - B. Regression
   - C. Recommendation
   - D. Anomaly detection
6. Which pair is correctly matched?
   - A. Customer groups → classification
   - B. Exam mark out of 100 → regression
   - C. Pass or fail → regression
   - D. Spam or not → clustering

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